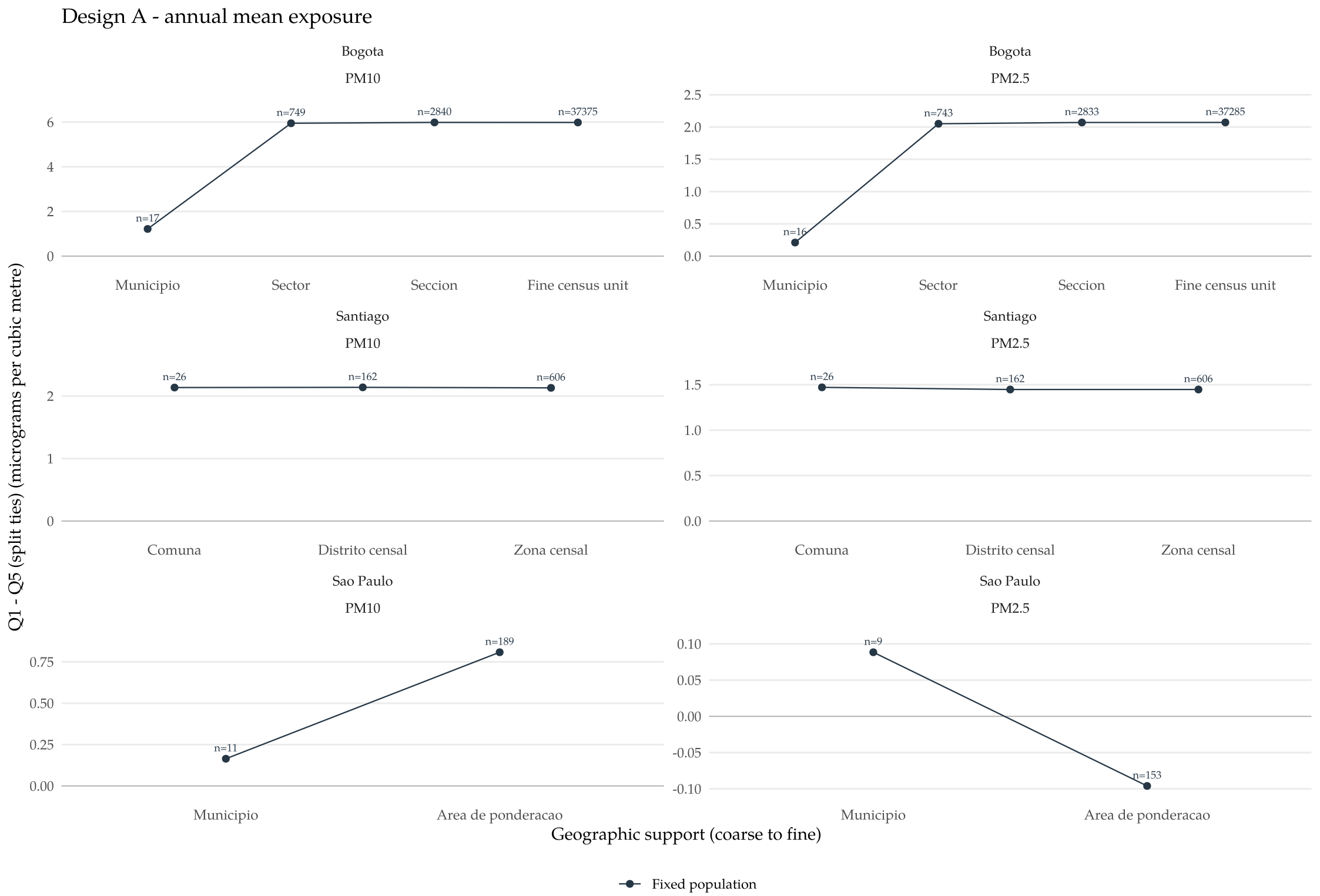
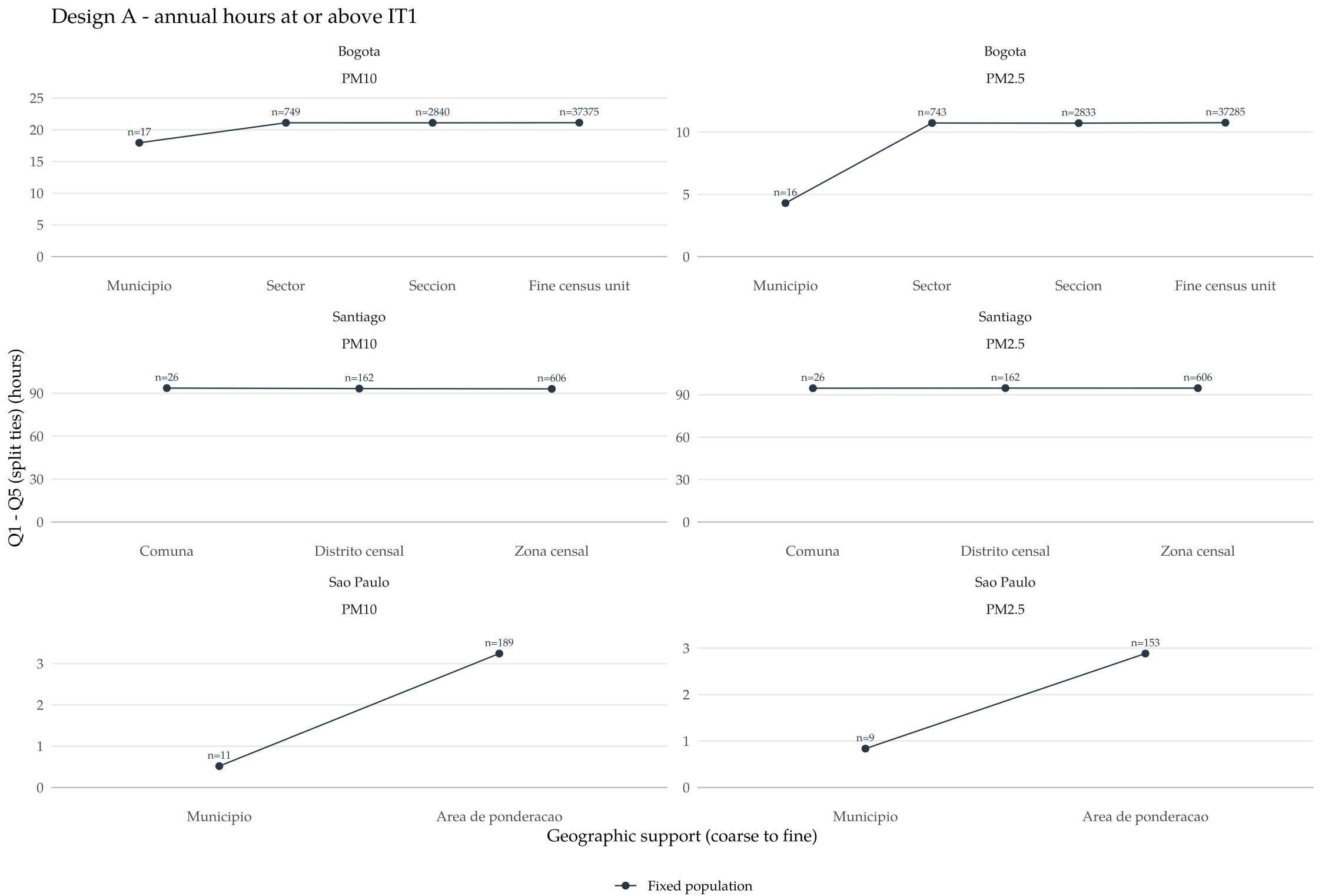
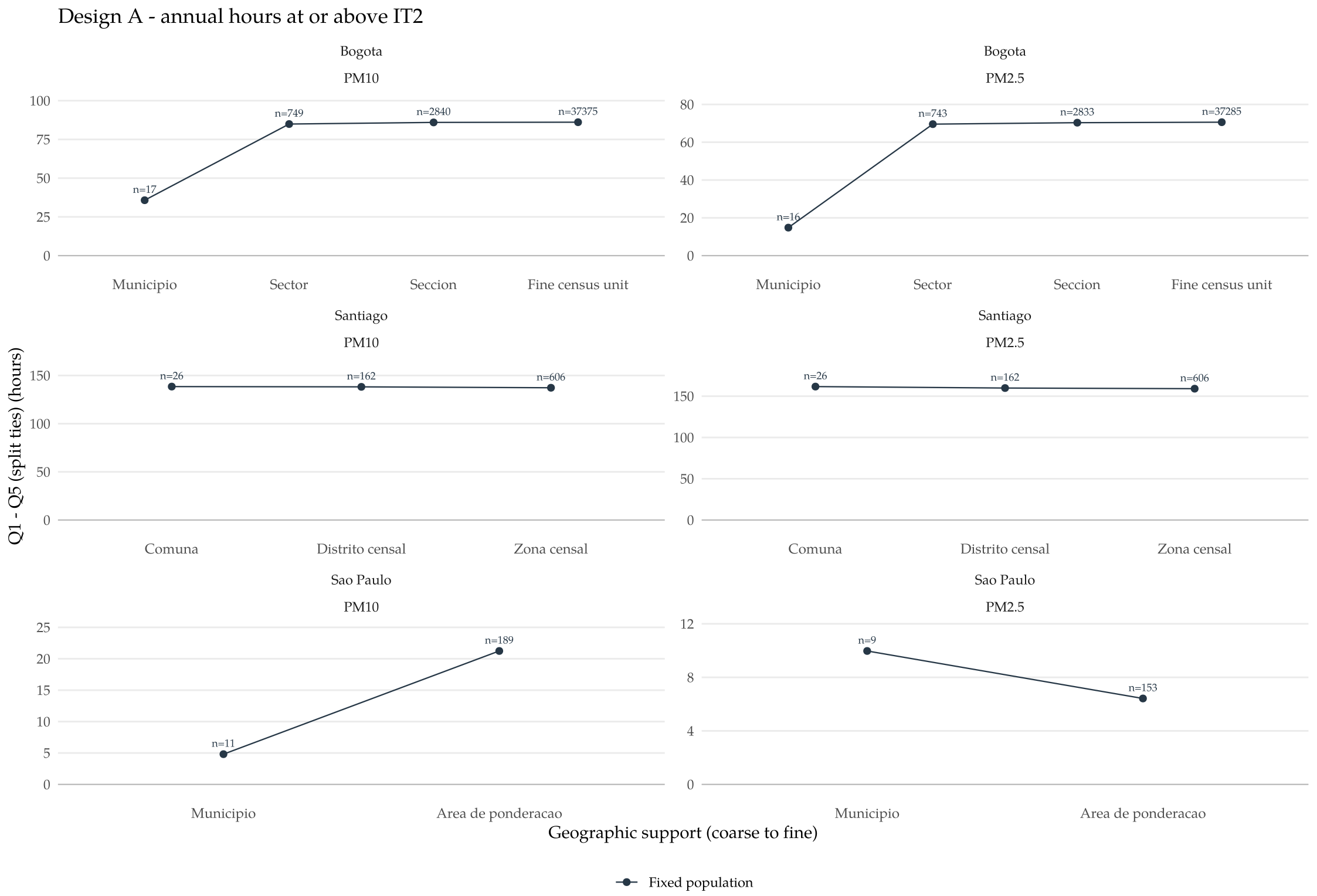


Fixed education-reporting adults and individual quintiles. Exposure aggregated with all-adult weights; locality branch shown separately.

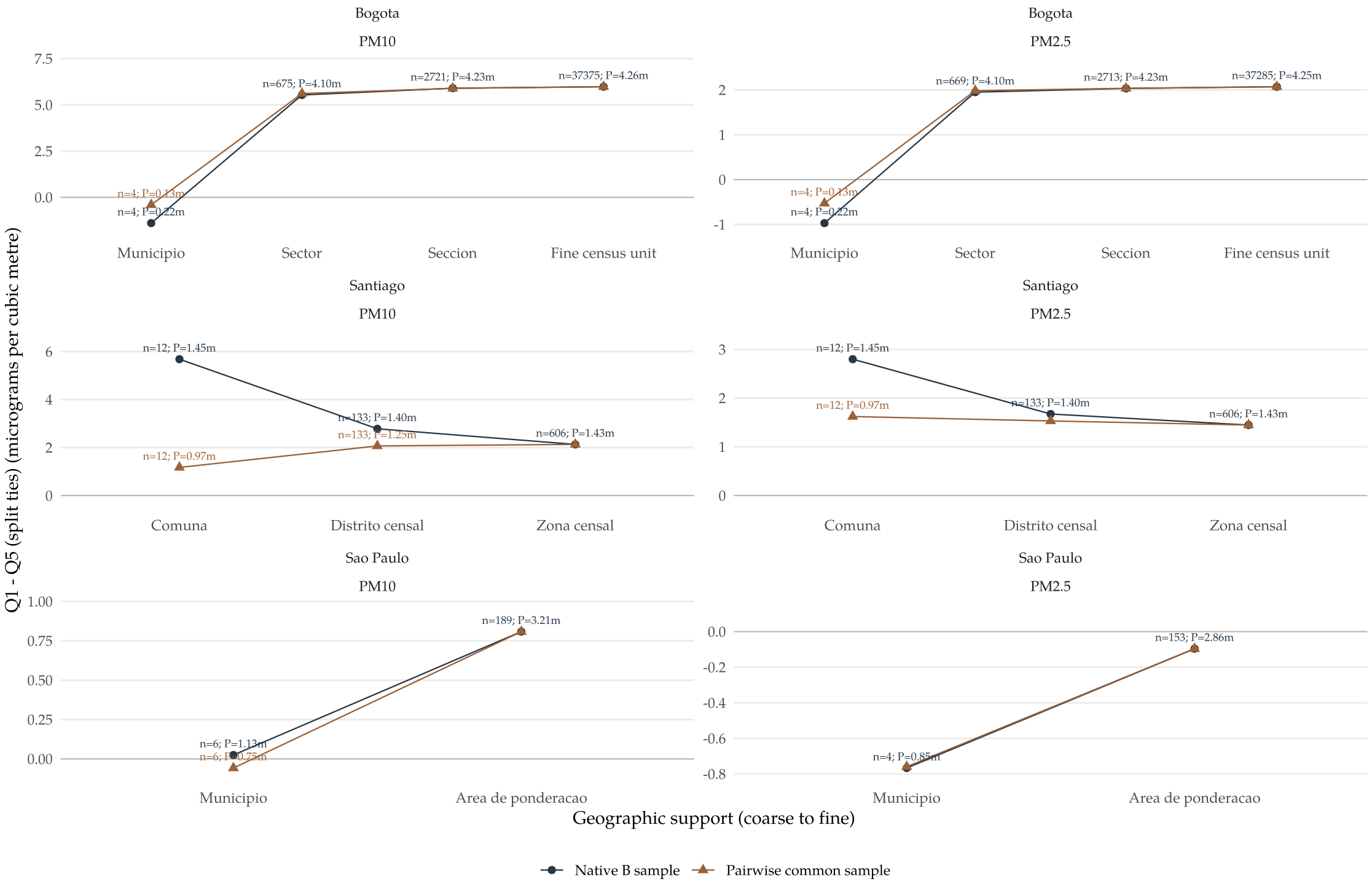


Fixed education-reporting adults and individual quintiles. Exposure aggregated with all-adult weights; locality branch shown separately.



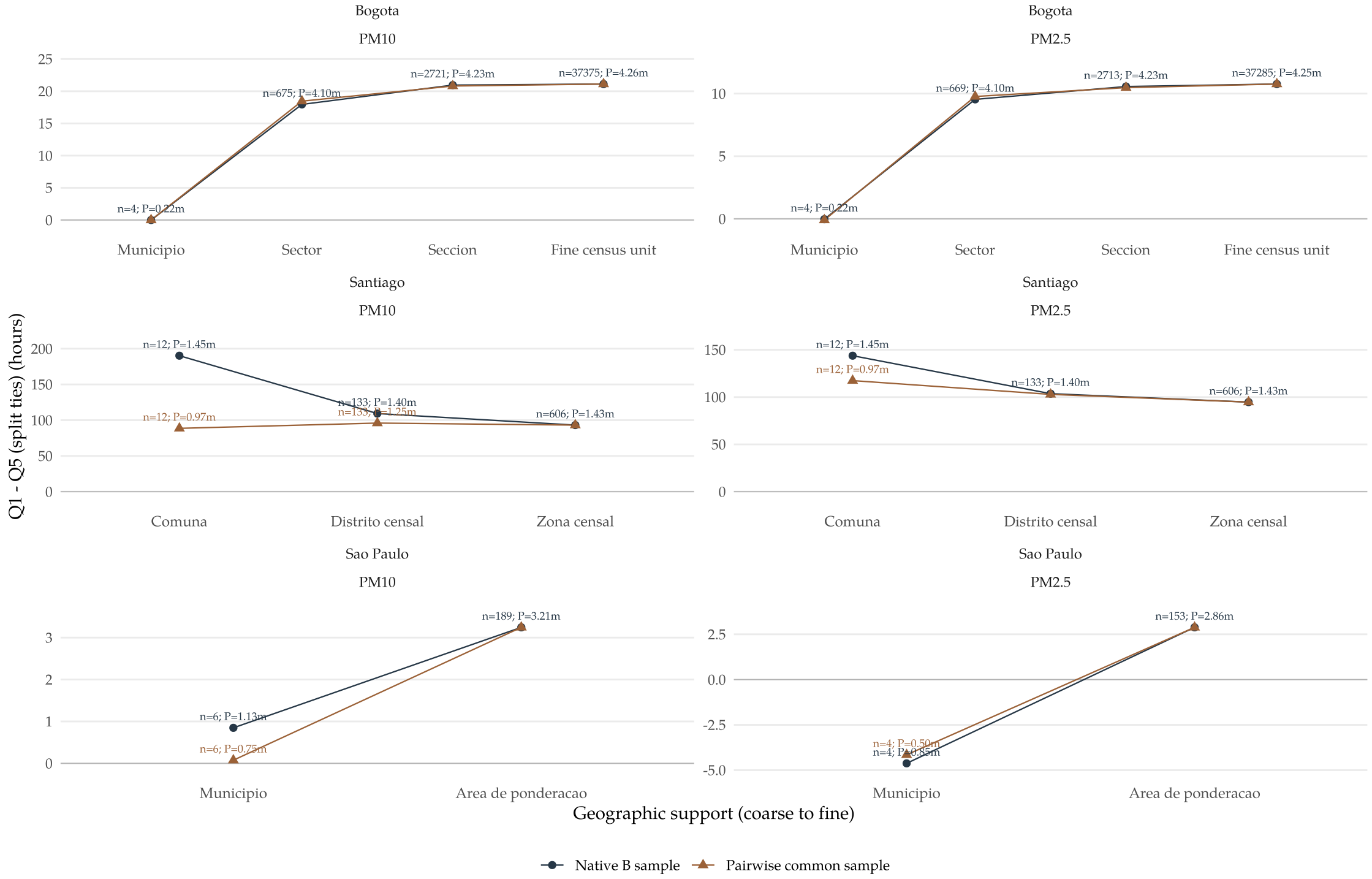
Fixed education-reporting adults and individual quintiles. Exposure aggregated with all-adult weights; locality branch shown separately.

Design B - annual mean exposure



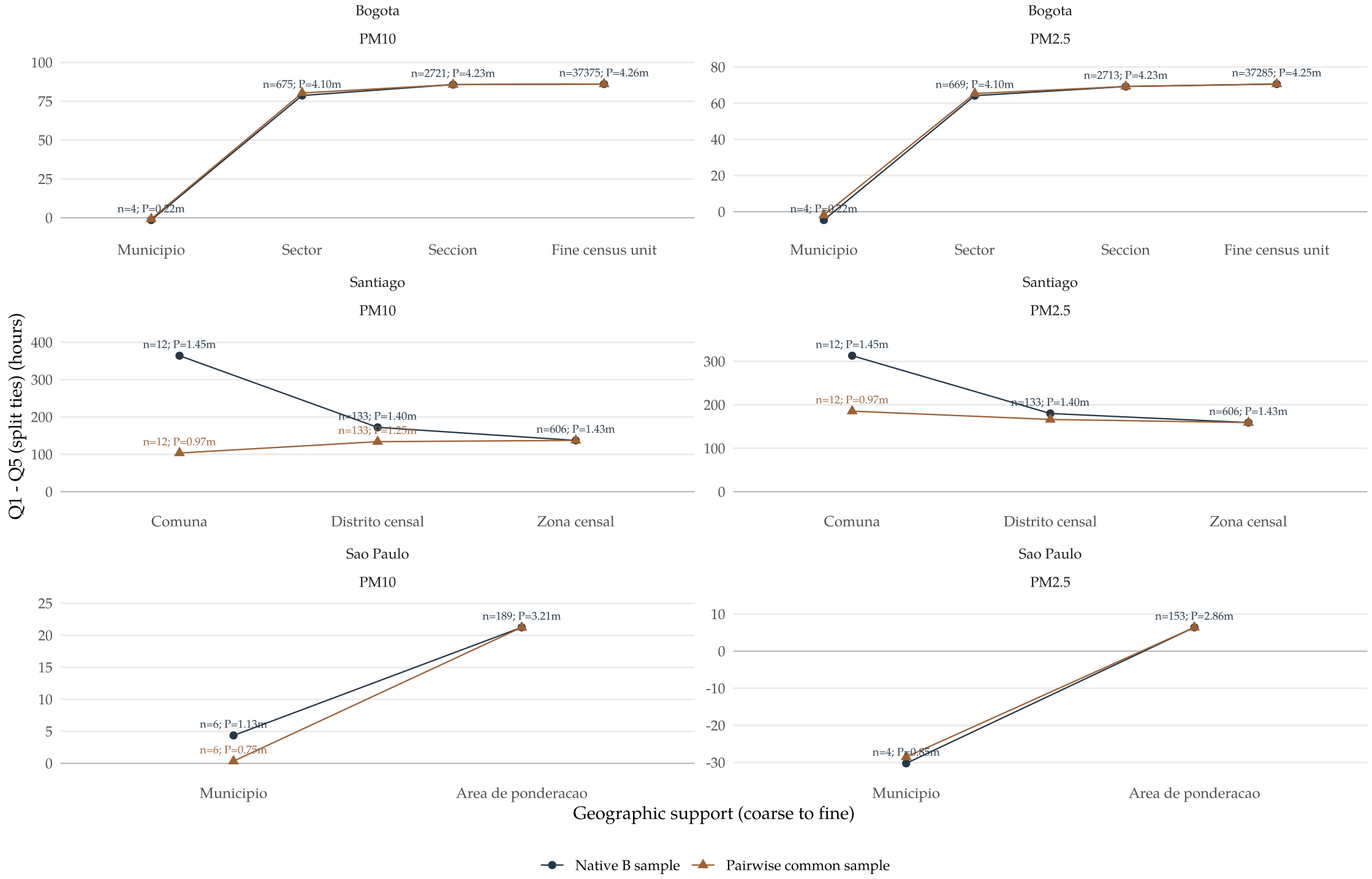
Native and common samples vary by level; P = education-reporting adults.
Common samples are pairwise intersections. Paired changes are shown separately.

Design B - annual hours at or above IT1

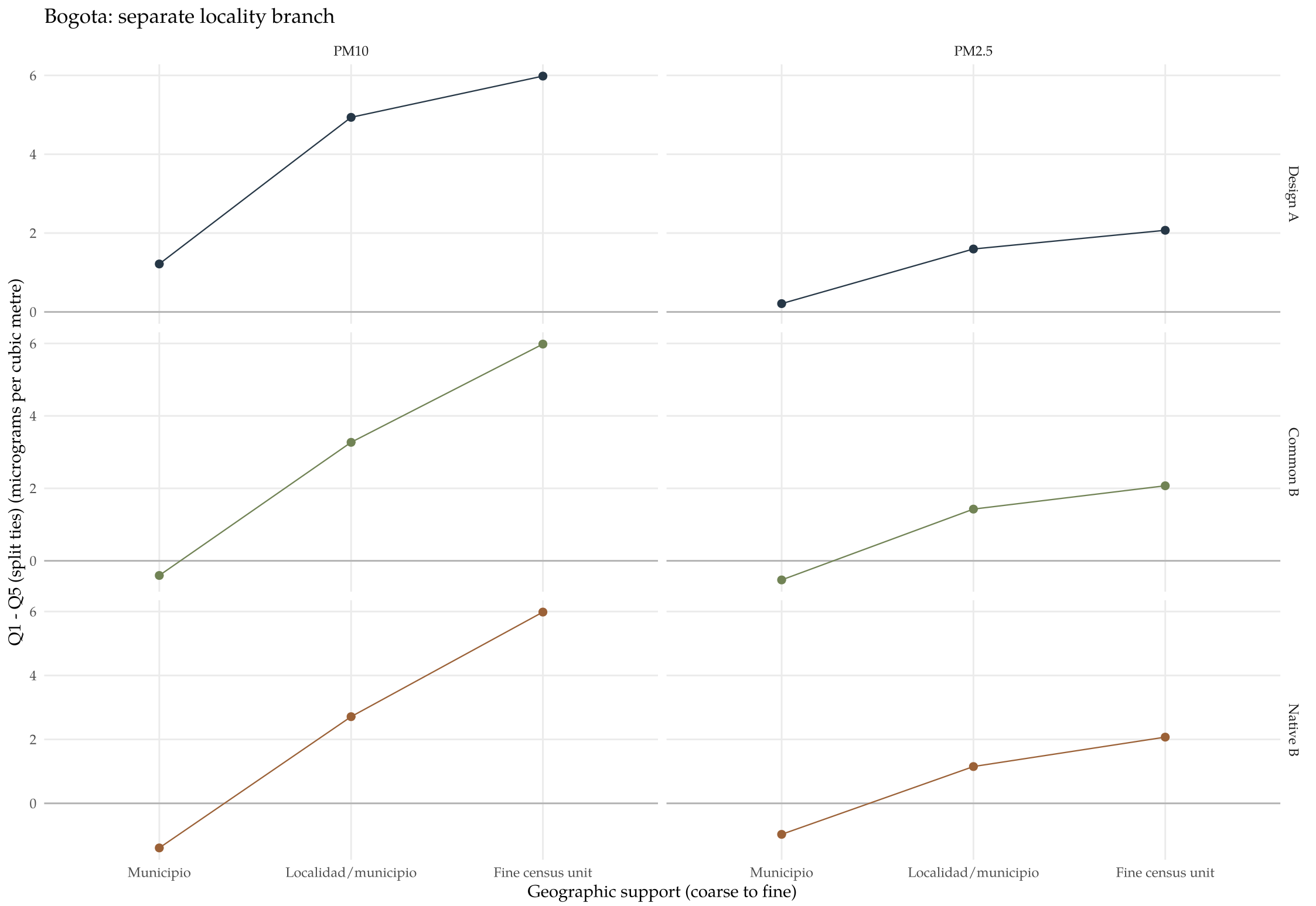


Native and common samples vary by level; P = education-reporting adults.
Common samples are pairwise intersections. Paired changes are shown separately.

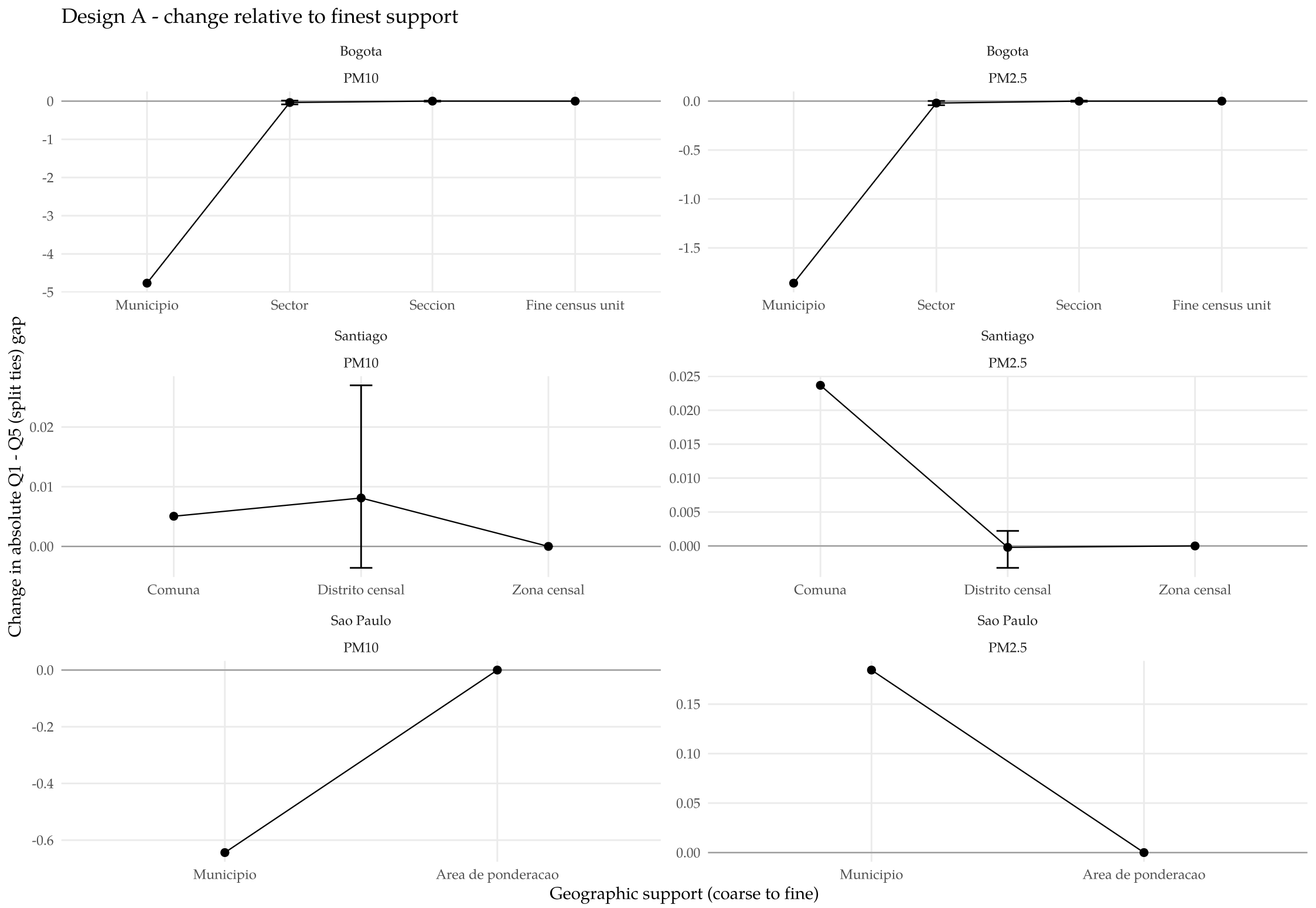
Design B - annual hours at or above IT2



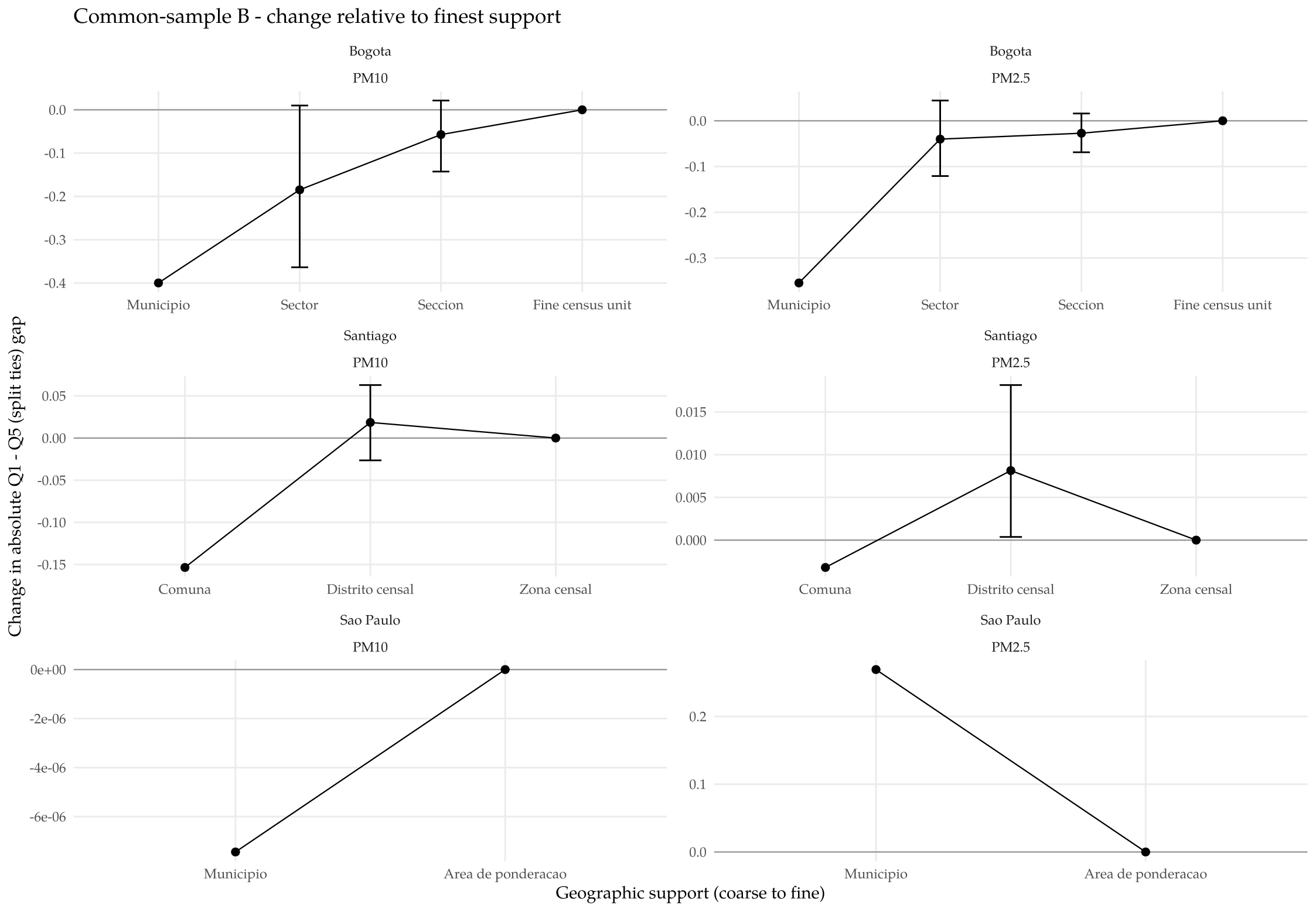
Native and common samples vary by level; P = education-reporting adults.
Common samples are pairwise intersections. Paired changes are shown separately.

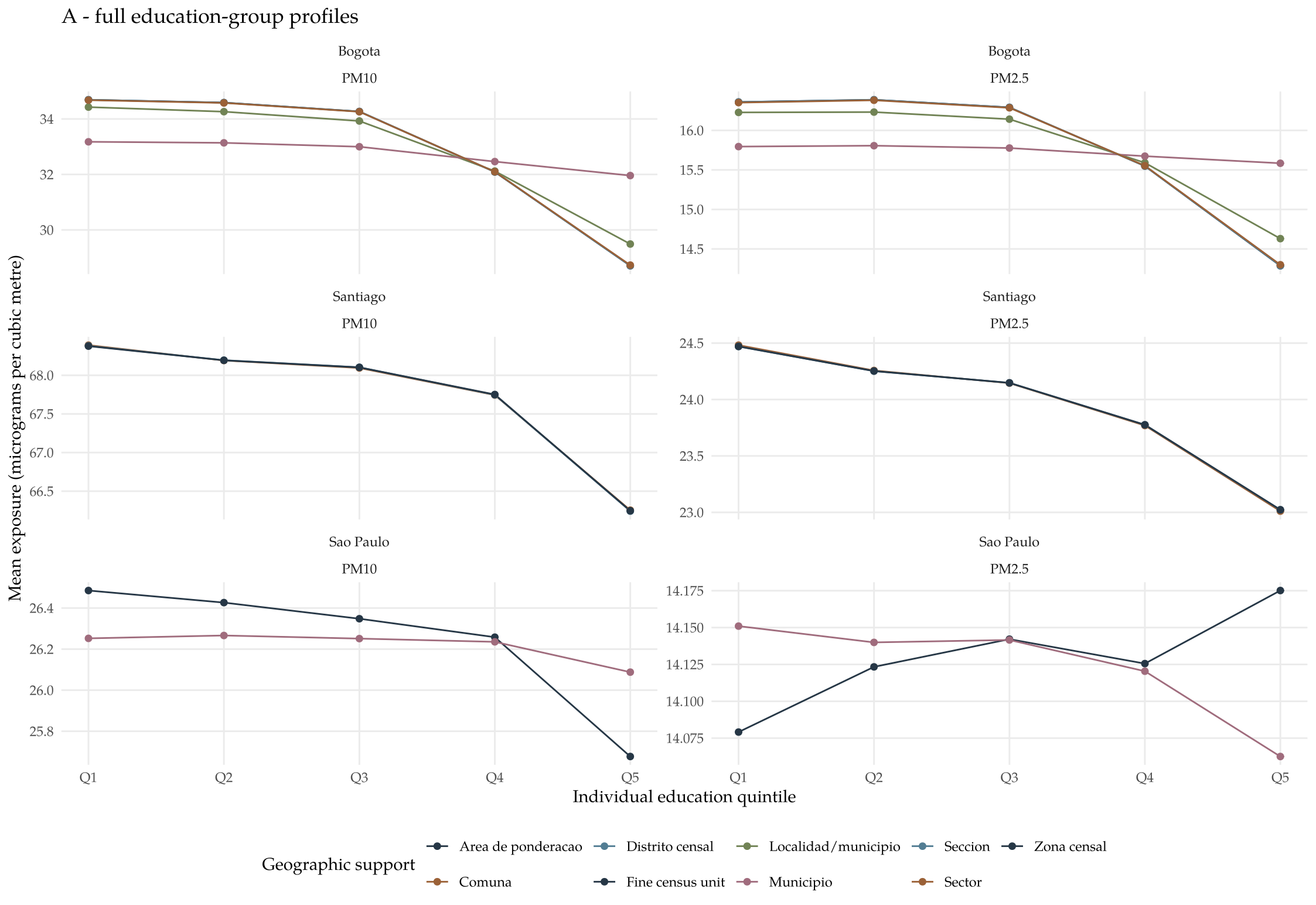


Descriptive comparisons.



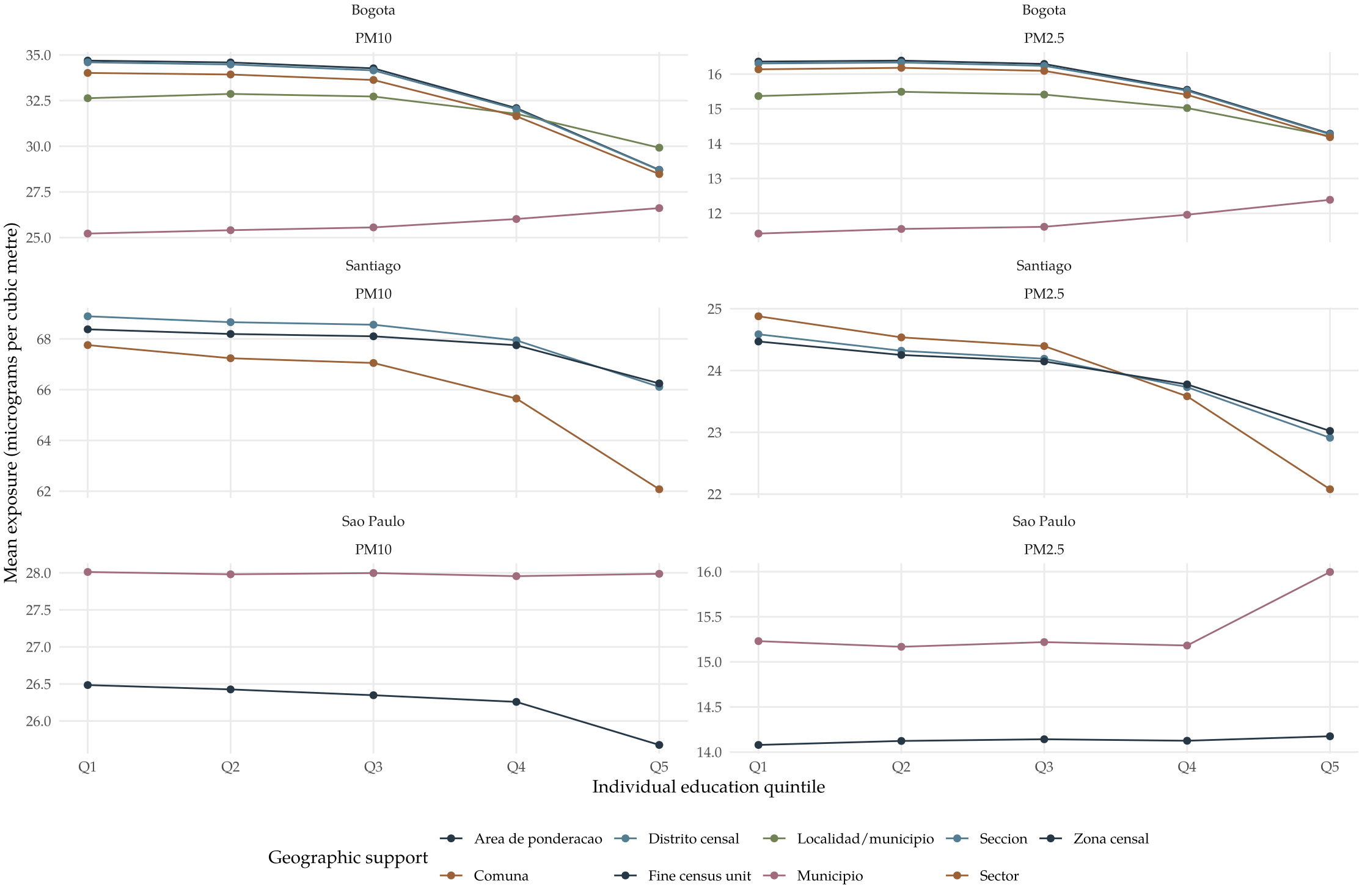
95% conditional geographic-cluster bootstrap intervals; 999 draws. Coarse comparisons are descriptive. B endpoints use the same pairwise population.


Common-sample B - change relative to finest support
Bogota
PM10
Bogota
PM2.5
Santiago
PM10
Santiago
PM2.5
Sao Paulo
PM10
Sao Paulo
PM2.5
Municipio
Sector
Seccion
Fine census unit
Comuna
Distrito censal
Zona censal
Area de ponderacao
Change in absolute Q1 - Q5 (split ties) gap
Geographic support (coarse to fine)
95% conditional geographic-cluster bootstrap intervals; 999 draws. Coarse comparisons are descriptive. B endpoints use the same pairwise population.



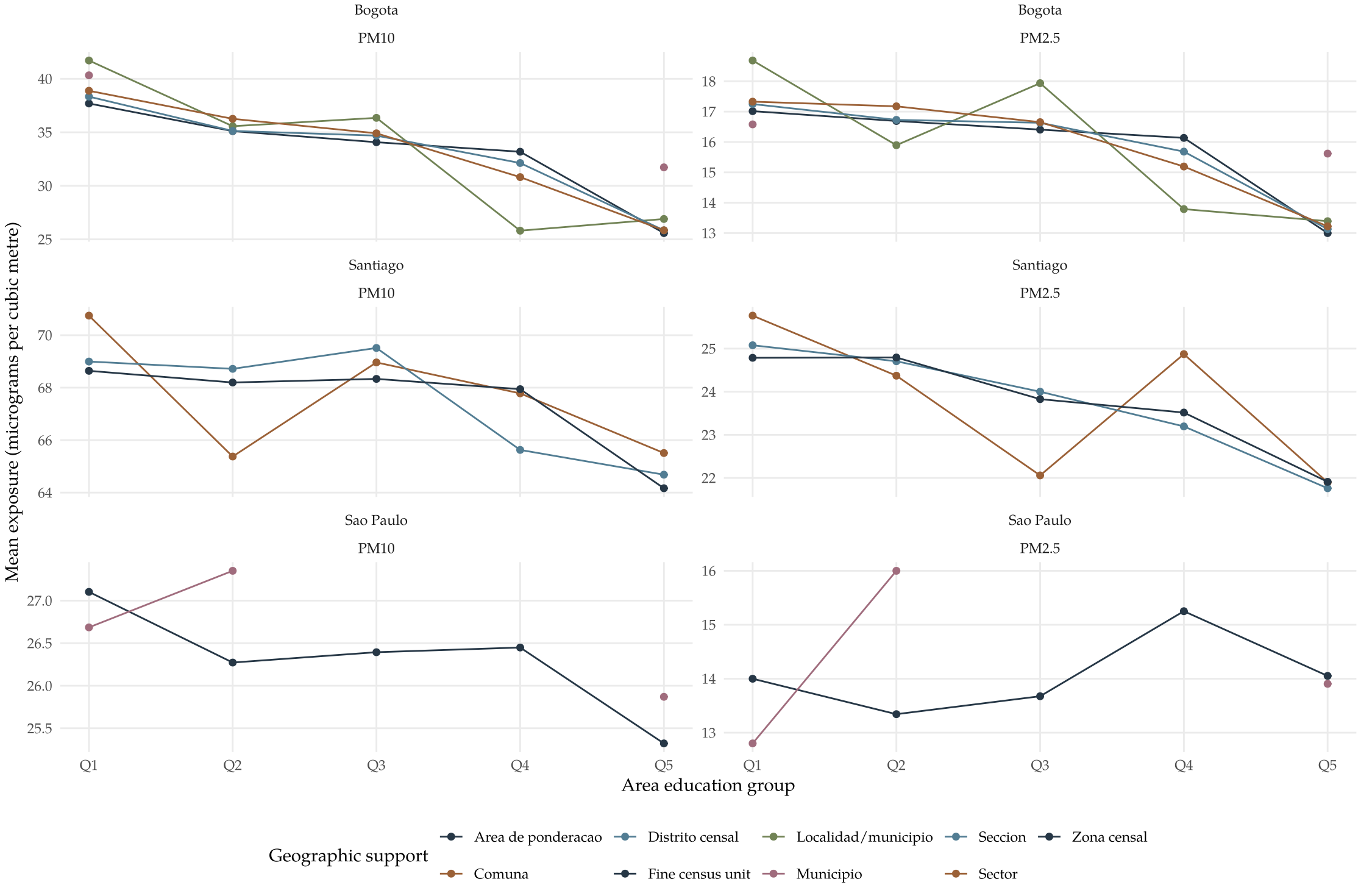
Quintiles with tied schooling values split proportionally across adjacent quintiles; group means use person weights.

Native B - full education-group profiles



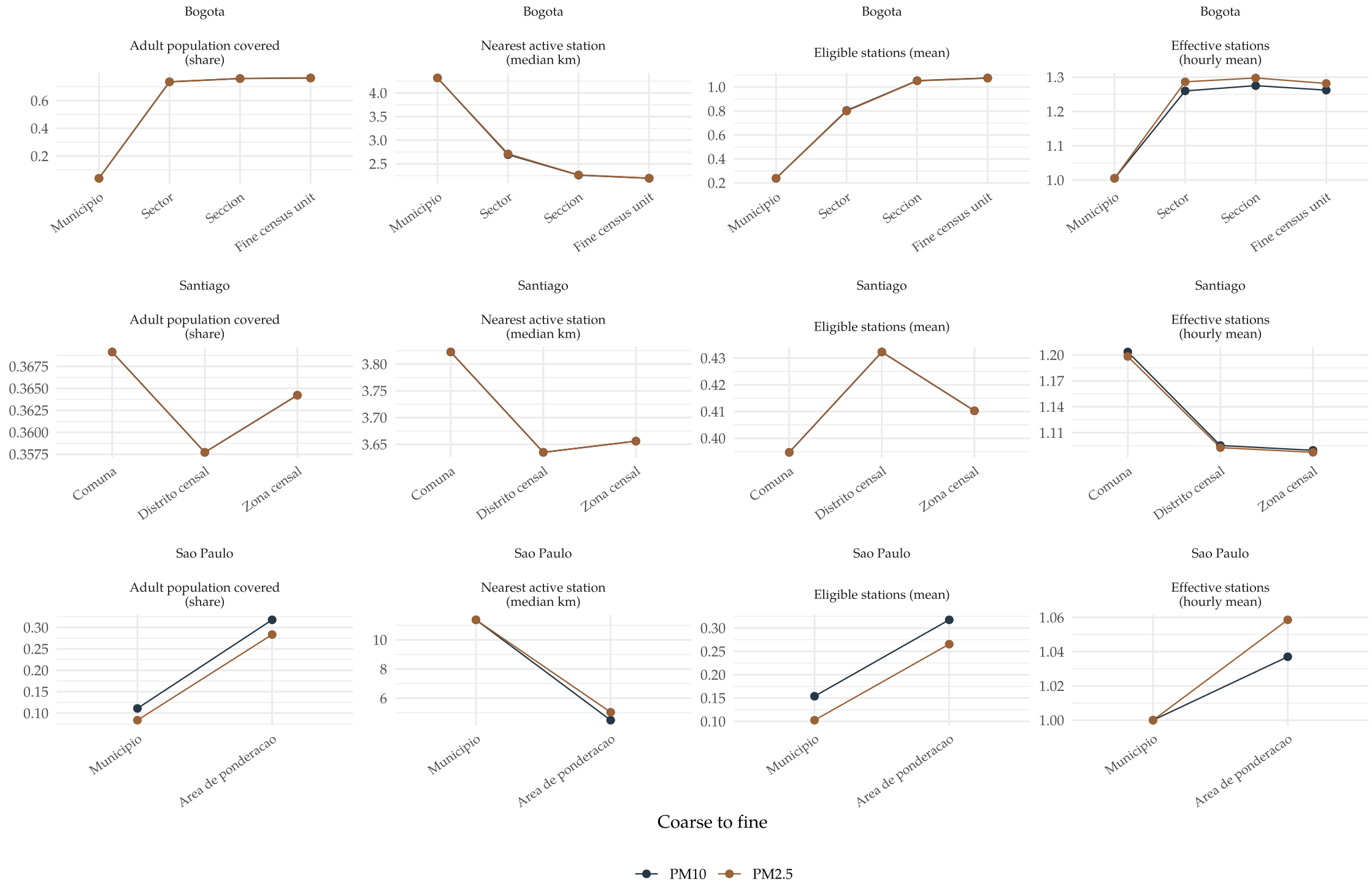
Quintiles with tied schooling values split proportionally across adjacent quintiles; group means use person weights.

C - full education-group profiles



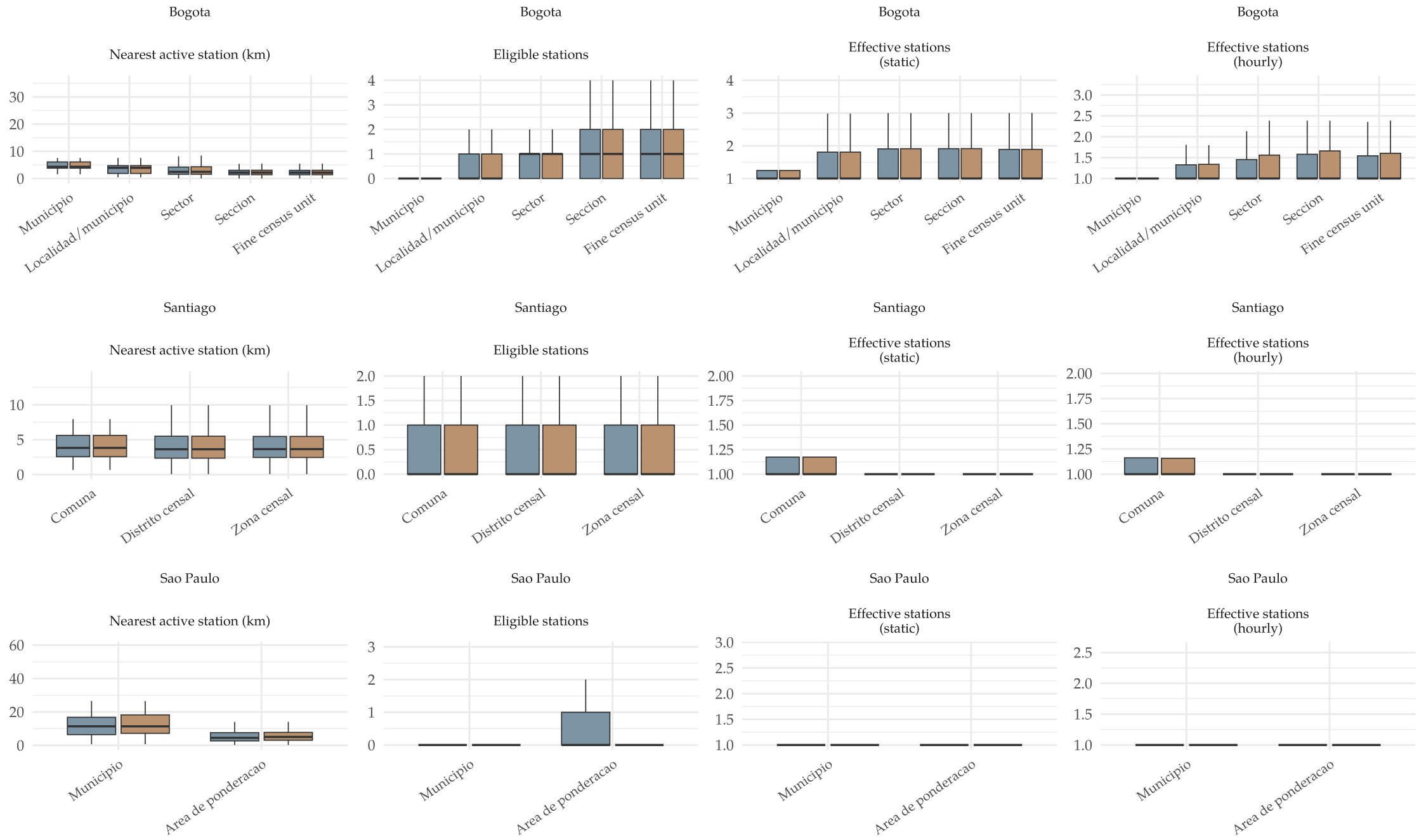
Classification only; finest exposure fixed. Empty area groups remain unavailable. These groups differ from A/B quintiles.

Monitoring-matrix sensitivity under Design B



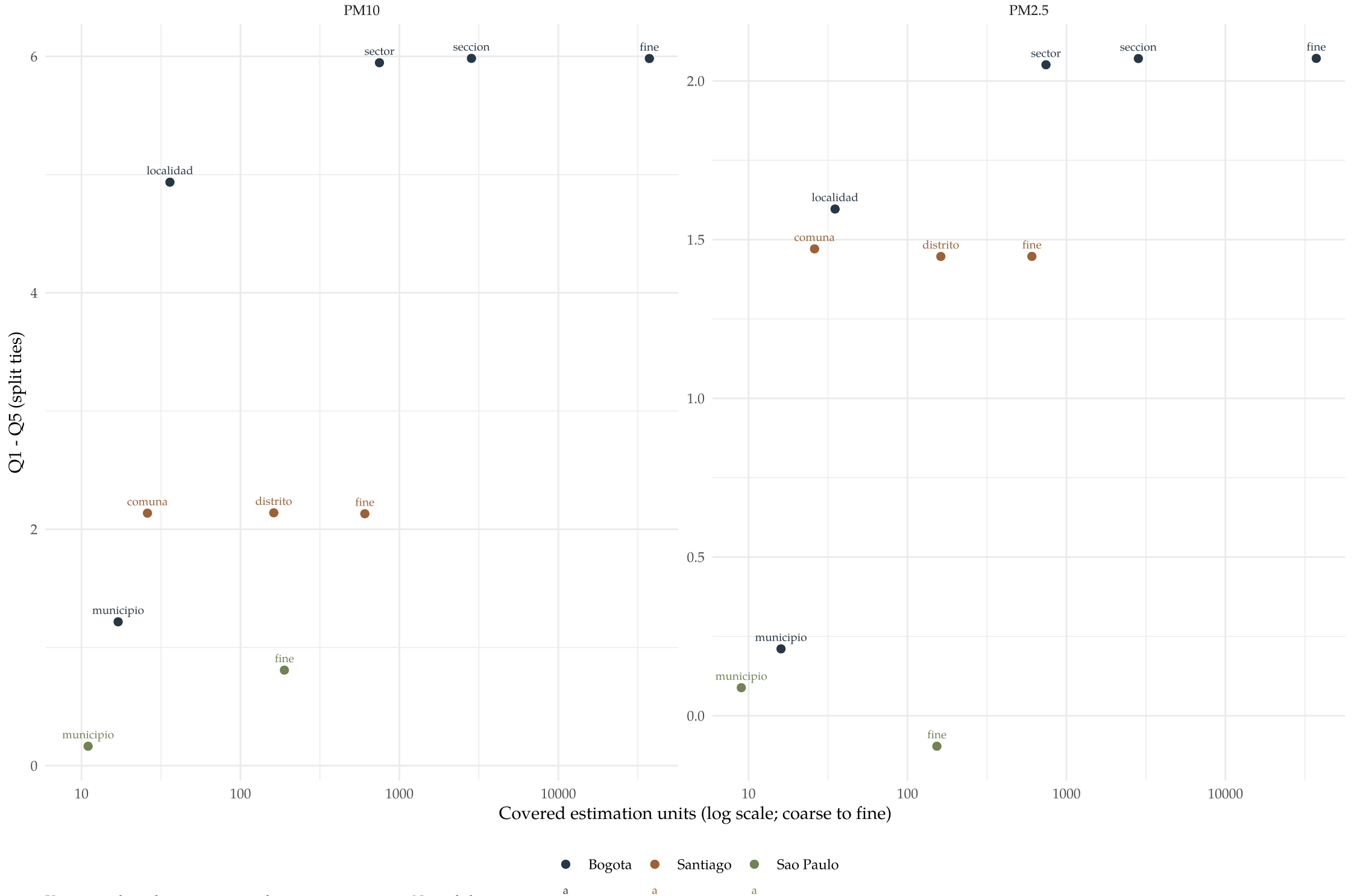
Catalog, active, eligible and contributing station counts are distinct in the tables.

Monitoring-matrix distributions across populated units

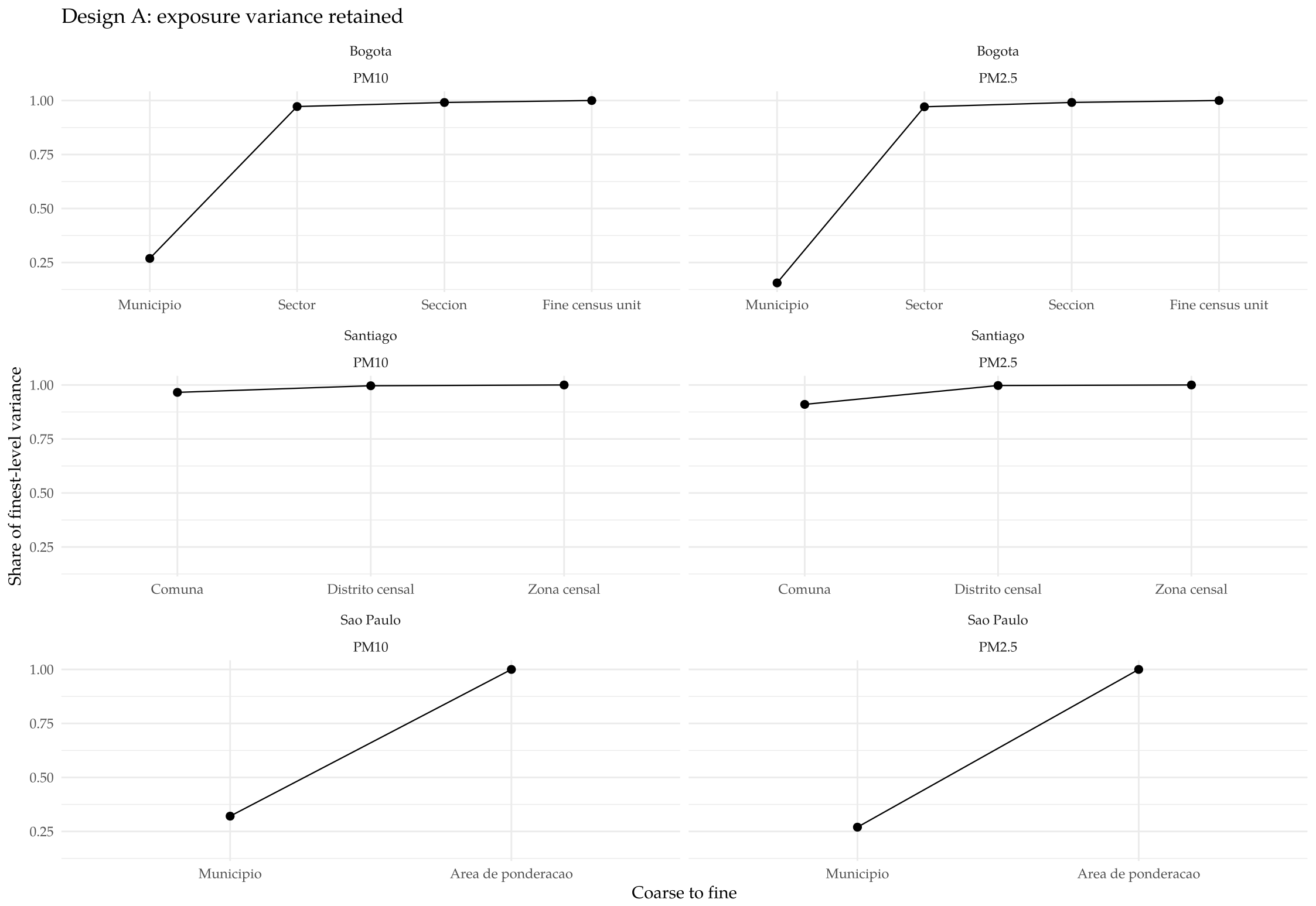


Boxes: unweighted unit quartiles; whiskers: 1.5 IQR. Outliers omitted from view only.

Design A: covered-unit counts as a supplementary resolution measure



Unit counts do not harmonize geographic concepts across cities. No pooled estimate.



All-adult aggregation weights and fixed covered support. Declining variance does not imply a monotone gap.